Riatiriba TVC

Voucher System

API Reference & Data Guide

Base URL: http://localhost:3000/api/vouchers

All request & response bodies are JSON | PDF endpoints stream binary

1. Data Shape — Voucher Object

Every API response that returns a voucher returns the same full object shape below. The list endpoint returns a subset of these fields (marked ✓ in the "In List?" column).

Field	Type	In List?	Description
id	integer	✓	Auto-increment primary key
voucherNo	string	✓	Serial number e.g. rtvc/v/0001 — assigned by server, never send this
date	string	✓	Voucher date — ISO 8601 format: YYYY-MM-DD
payeeName	string	✓	Full name of the payee
payeeAddress	string	—	Physical or postal address of payee
accountName	string	—	Bank / account name for payment
lineItems	LineItem[]	—	Array of particulars rows — see sub-type below
amountInWords	string	✓	Auto-computed by server from lineItems — do NOT send
chequeNo	string	✓	Cheque number for this payment
voteheadItems	VHItem[]	—	Array of expenditure votehead rows — see sub-type below
preparedBy	string	—	Name of staff who prepared the voucher
preparedDate	string	—	ISO date when voucher was prepared
approvedBy	string	—	Finance Officer name
approvedDate	string	—	ISO date of finance officer approval
receivedSum	string	—	Text description of sum received (filled after signing)
receivedCashOrCheque	string	—	"Cash" or "Cheque" — filled after signing
receiverSignature	string	—	Name/reference for receiver signature line
receiverIdNo	string	—	National ID number of receiver
receivedDate	string	—	ISO date voucher was received / signed
authorizedByPrincipal	string	—	Principal name who authorized
authorizedDate	string	—	ISO date of principal authorization
pdfGenerated	boolean	—	True once PDF has been streamed at least once

Field	Type	In List?	Description
createdAt	string	✓	ISO timestamp of record creation
updatedAt	string	—	ISO timestamp of last update

LineItem sub-type (used in lineItems[ ])

Field	Type	Required	Description
date	string	No	Row date — free text or ISO date
particulars	string	Yes	Description of the expense / service
kshs	number	No	Amount in Kenya Shillings (integer part)
cts	number	No	Cents portion 0–99. Values >99 roll into shillings automatically

VHItem sub-type (used in voteheadItems[ ])

Field	Type	Required	Description
date	string	No	Row date — free text or ISO date
expenditureVotehead	string	Yes	Votehead name / code e.g. "Office Supplies (1200)"
kshs	number	No	Amount in Kenya Shillings
cts	number	No	Cents portion 0–99

2. Routes

All routes are mounted at /api/vouchers. Successful responses are 200 OK or 201 Created. Errors always return JSON with an "error" key.

2.1 List Vouchers

`GET` **`/api/vouchers`** — Returns a summary list of all vouchers, newest first

No request body required.

Response — 200 OK

```
[
  {
    "id": 3,
    "voucherNo": "rtvc/v/0003",
    "date": "2025-06-21",
    "payeeName": "John Kariuki Mwangi",
    "amountInWords": "Kenya Shillings Twenty Five Thousand Only",
    "chequeNo": "CHQ-00451",
    "createdAt": "2025-06-21T10:32:00.000Z"
  },
  { ... }]
```

2.2 Get Single Voucher

`GET` **`/api/vouchers/:id`** — Returns the full voucher object for the given id

No request body required. :id is the integer primary key.

Response — 200 OK

```
{
  "id": 3,
  "voucherNo": "rtvc/v/0003",
  "date": "2025-06-21",
  "payeeName": "John Kariuki Mwangi",
  "payeeAddress": "P.O. Box 100, Gitaru Town",
  "accountName": "Operations Account",
  "lineItems": [
    { "date": "2025-06-21", "particulars": "Supply of stationery", "kshs": 20000, "cts": 0 }
  ],
  "amountInWords": "Kenya Shillings Twenty Thousand Only",
  "chequeNo": "CHQ-00451",
  "voteheadItems": [
    { "date": "2025-06-21", "expenditureVotehead": "Office Supplies (1200)", "kshs": 20000,
"cts": 0 }
  ],
  "preparedBy": "Jane Otieno",
  "preparedDate": "2025-06-21",
  "approvedBy": "Samuel Waweru",
  "approvedDate": "2025-06-21",
  "receivedSum": null,
  "receivedCashOrCheque": null,
  "receiverSignature": null,
  "receiverIdNo": null,
  "receivedDate": null,
  "authorizedByPrincipal": null,
  "authorizedDate": null,
  "pdfGenerated": false,
  "createdAt": "2025-06-21T10:32:00.000Z",
  "updatedAt": "2025-06-21T10:32:00.000Z"}
```


Response — 404 Not Found


```
{ "error": "Not found" }
```


2.3 Create Voucher

POST **/api/vouchers** — Creates one voucher. Server assigns voucherNo and computes amountInWords.

> *Do NOT include voucherNo or amountInWords in the request — both are computed server-side and any values sent will be ignored.*

Request Body


```
{
  "date": "2025-06-21",
  "payeeName": "John Kariuki Mwangi",
  "payeeAddress": "P.O. Box 100, Gitaru Town",
  "accountName": "Operations Account",
  "lineItems": [
    { "date": "2025-06-21", "particulars": "Supply of stationery items", "kshs": 12000,
"cts": 50 },
    { "date": "2025-06-21", "particulars": "Printing services",          "kshs": 8000,
"cts": 0  }
  ],
  "chequeNo": "CHQ-00451",
  "voteheadItems": [
    { "date": "2025-06-21", "expenditureVotehead": "Office Supplies (1200)",       "kshs":
12000, "cts": 50 },
    { "date": "2025-06-21", "expenditureVotehead": "Printing & Stationery (1201)", "kshs":
8000,  "cts": 0  }
  ],
  "preparedBy": "Jane Otieno",
  "preparedDate": "2025-06-21",
  "approvedBy": "Samuel Waweru",
  "approvedDate": "2025-06-21"}
```


Response — 201 Created

Returns the full voucher object (same shape as GET /api/vouchers/:id) with server-assigned voucherNo and computed amountInWords.

```
{
  "id": 4,
  "voucherNo": "rtvc/v/0004",
  "amountInWords": "Kenya Shillings Twenty Thousand and Fifty Cents Only",
  ... (all other fields)}
```

Response — 500 Internal Server Error

```
{ "error": "<sequelize or db error message>" }
```

2.4 Batch Create Vouchers

POST **/api/vouchers/batch** — Creates N blank vouchers with sequential serial numbers

> *Useful for pre-printing a book of blank serialised vouchers. All records share the same template fields. Max 200 per call.*

Request Body

```
{
  "count": 10,
  "template": {
    "date": "2025-06-21",
    "preparedBy": "Jane Otieno",
    "approvedBy": "Samuel Waweru"
  }}
```

> *"template" is optional — omit it to generate fully blank vouchers. Only count is required.*

Response — 201 Created

```
{
  "created": 10,
  "vouchers": [
    { "id": 5,  "voucherNo": "rtvc/v/0005" },
    { "id": 6,  "voucherNo": "rtvc/v/0006" },
    ...
    { "id": 14, "voucherNo": "rtvc/v/0014" }
  ]}
```

Response — 400 Bad Request

```
{ "error": "count must be between 1 and 200" }
```

2.5 Update Voucher

PUT **/api/vouchers/:id** — Updates any fields on an existing voucher. Recomputes amountInWords if lineItems change.

> *Send only the fields you want to change — you do not need to send the full object. voucherNo cannot be changed. amountInWords is always recomputed from lineItems.*

Request Body (partial update example)

```
{
  "chequeNo": "CHQ-00455",
  "authorizedByPrincipal": "Dr. Alice Muthoni",
  "authorizedDate": "2025-06-22",
  "receivedCashOrCheque": "Cheque",
  "receiverIdNo": "12345678",
  "receivedDate": "2025-06-22"}
```

Response — 200 OK

Returns the full updated voucher object.

Response — 404 Not Found

```
{ "error": "Not found" }
```

2.6 Delete Voucher

DELETE **/api/vouchers/:id** — Permanently deletes the voucher record

Response — 200 OK

```
{ "message": "Deleted" }
```

Response — 404 Not Found

```
{ "error": "Not found" }
```

2.7 Download Single Voucher PDF

GET **/api/vouchers/:id/pdf** — Streams a single-page A4 PDF for the given voucher

Response Content-Type is application/pdf. Open in a new browser tab or use as blob URL. After streaming, sets pdfGenerated = true on the record.

Response Headers

```
Content-Type: application/pdfContent-Disposition: inline; filename="rtvc/v/0003.pdf"
```

Fetch example

```
// Open PDF directly in a new tab
window.open(`/api/vouchers/${id}/pdf`, "_blank");

// Or get as blob for preview
const res  = await fetch(`/api/vouchers/${id}/pdf`);
const blob = await res.blob();
const url  = URL.createObjectURL(blob);window.open(url, "_blank");
```

Response — 404 Not Found

```
{ "error": "Not found" }
```

2.8 Download Batch PDF

POST **/api/vouchers/batch-pdf** — Streams a multi-page PDF — one voucher per page

> *Use ids[] to select specific vouchers, or all: true to include every voucher in the DB. Returns a single PDF file.*

Request Body — specific IDs

```
{ "ids": [1, 2, 5, 9] }
```

Request Body — all vouchers

```
{ "all": true }
```

Response Headers

```
Content-Type: application/pdfContent-Disposition: inline; filename="vouchers-batch-
1719043200000.pdf"
```

Fetch example

```
const res  = await fetch("/api/vouchers/batch-pdf", {
  method: "POST",
  headers: { "Content-Type": "application/json" },
  body: JSON.stringify({ ids: [1, 2, 3] }),
```

```
});
const blob = await res.blob();window.open(URL.createObjectURL(blob), "_blank");
```

Response — 400 Bad Request

```
{ "error": "Provide ids[] or all:true" }
```

Response — 404 Not Found

```
{ "error": "No vouchers found" }
```

3. Error Shapes

All errors return JSON. The frontend should always check res.ok before parsing the body as a voucher.

HTTP Status	When	"error" value
400 Bad Request	Validation failed (e.g. bad count)	Human-readable description of the problem
404 Not Found	No voucher with the given :id	`"Not found"` or `"No vouchers found"`
500 Internal Server Error	DB error, serialization error, file I/O	Sequelize or Node.js error message — log it

Recommended frontend error handler

```
async function apiFetch(url, options = {}) {
  const res  = await fetch(url, {
    headers: { "Content-Type": "application/json" },
    ...options,
  });
  const data = await res.json();
  if (!res.ok) throw new Error(data.error || `HTTP ${res.status}`);
  return data;
}

// Usage
try {
  const voucher = await apiFetch("/api/vouchers/42");
} catch (err) {
  showToast(err.message);   // e.g. "Not found"}
```

4. Quick Reference

Method	Path	Body	Returns
GET	/api/vouchers	—	Voucher[] (summary)
GET	/api/vouchers/:id	—	Voucher (full)
POST	/api/vouchers	Voucher fields	Voucher (full) — 201
POST	/api/vouchers/batch	{ count, template }	{ created, vouchers[] } — 201
PUT	/api/vouchers/:id	Partial fields	Voucher (full) — 200
DELETE	/api/vouchers/:id	—	{ message } — 200
GET	/api/vouchers/:id/pdf	—	PDF binary stream
POST	/api/vouchers/batch-pdf	{ ids } or { all }	PDF binary stream (multi-page)

Fields the server always computes — never send these

Field	Computed from	Notes
voucherNo	DB sequence + config/voucherSerial.js	Format: rtvc/v/0001. Change format in that config file.
amountInWords	lineItems kshs + cts totals	Uses number-to-words. Cts >99 roll into shillings.
id	Auto-increment	Use this as the key for GET/PUT/DELETE/:id
createdAt	Sequelize timestamps	ISO 8601 UTC string
updatedAt	Sequelize timestamps	ISO 8601 UTC string

5. Serial Number Configuration

To change the voucher serial format, edit backend/config/voucherSerial.js only — no other file needs to change.

Constant	Default	Effect
SERIAL_PREFIX	`"rtvc"`	The institution code at the start of every serial
SERIAL_MIDDLE	`"v"`	Middle segment (e.g. document type). Set to '' to omit entirely
SERIAL_START	1	First integer sequence number — renders as 0001 due to padding
SERIAL_PADDING	4	Zero-pad width. 4 → 0001, 5 → 00001
SERIAL_SEP	`"/"`	Separator between parts. Use "-" to get rtvc-v-0001

Format examples

PREFIX	MIDDLE	START	SEP	Result
`"rtvc"`	`"v"`	1	`"/"`	rtvc/v/0001 ← current default
`"RTVC"`	`"PV"`	1	`"-"`	RTVC-PV-0001
`"rtvc"`	`""`	1	`"/"`	rtvc/0001
`"rtvc"`	`"v"`	500	`"/"`	rtvc/v/0500 (start from 500)